

# ROTATIONAL DYNAMICS



“ALL THAT IS IMPORTANT IN THIS  
ONE MOMENT IS MOVEMENT”

## THEORY ASSIGNMENT

1. Discuss the necessity of radius of gyration. Define it. On what factors does it depend and it does not depend? Can you locate some similarity between the centre of mass and radius of gyration?
2. On what factors does the frequency of a conical pendulum depends? Is it independent of some factors?
3. Why are curved roads banked?
4. Derive an expression for time period of a conical pendulum.
5. State and prove the parallel axis theorem.

## NUMERICALS

1. A ceiling fan having moment of inertia  $2 \text{ kg-m}^2$  attains its maximum frequency of 60 rpm in ' $2\pi$ ' seconds. Calculate its power rating.
2. A racing track of curvature 9.9 m is banked at  $\tan^{-1} 0.5$ . Coefficient of static friction between the track and the tyres of a vehicle is 0.2. Determine the speed limits with 10 % margin.
3. A fan is rotating at 90 rpm. It is then switched OFF. It stops after 21 revolutions. Calculate the time taken by it to stop assuming that the frictional torque is constant.
4. A spherical water balloon is revolving at 60 rpm. In the course of time, 48.8 % of its water leaks out. With what frequency will the remaining balloon revolve now? Neglect all non-conservative forces.